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# **The Influence of Mutation Bias on Adaptive Evolution**

BACHELOR THESIS

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**Title:** The Influence of Mutation Bias on Adaptive Evolution

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**Abstract:**

In evolutionary biology, natural selection acts upon genetic variation, which arises through mutations. However, mutations do not happen with equal probability, their occurrence is shaped by mutation biases; for example, a general tendency to favour transitions over transversions, or to prefer substitutions towards AT rather than GC. Recent studies clearly document that such biases are not overridden by natural selection but can influence the direction of adaptive evolution. Studies explored in this thesis have shown that the spectrum of adaptive substitutions can be predicted from the mutation spectrum, and even vice versa. Many underlying biases however remain unknown or unexplored. In this thesis, we investigate whether the structure of the genetic code introduces a mutation bias that influences the frequency of adaptive substitutions. We hypothesised that changes accessible through a greater number of single-nucleotide substitutions would occur more frequently among adaptive mutations. To test this, we applied correlation analyses and a negative binomial model to a dataset of antibiotics resistance-conferring substitutions in *Mycobacterium tuberculosis*. The analysis did not reveal a significant relationship, potentially due to disruptive factors in the data, such as their skewed distribution or other mutation biases.

**Keywords:** mutation bias, genetic code, adaptive evolution, *Mycobacterium tuberculosis*

**Název práce:** Vliv nerovnoměrné rychlosti vzniku mutací na adaptivní evoluci

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**Abstrakt:**

Základním mechanismem evoluční biologie je přirozený výběr, který vybírá nejvýhodnější varianty v rámci genetické variability, jež je tvořena mutacemi. Mutace však nevznikají zcela náhodně se stejnou pravděpodobností, naopak, jejich výskyt je ovlivněn tzv. mutačními biasy, tedy různými tendencemi mutačních mechanismů, například obecnou preferencí tran-zic před transverzemi nebo substitucemi na AT před GC. Současné studie demonstrují, že tyto nerovnoměrnosti nejsou přirozeným výběrem potlačeny, ale ovlivňují směr adaptivní evoluce. Články shrnuté v této bakalářské práci ukazují, že se z mutačního spektra dá vyvodit spektrum adaptivních substitucí, a dokonce i naopak. Mnoho mutačních biasů však zatím není objeveno či prozkoumáno. V této práci jsme si položily otázku, zda samotná struktura genetického kódu nezavádí další mutační tendenci, která ovlivňuje poměr adaptivních substitucí. Předpokládaly jsme, že změny dosažitelné větším počtem jedno-nukleotidových substitucí se budou mezi adaptivními mutacemi vyskytovat častěji. Tento předpoklad jsme testovaly pomocí korelační analýzy a negativně binomického modelu na datasetu substitucí vedoucích k rezistenci vůči antibiotikům u druhu *Mycobacterium tuberculosis*. Analýzy však žádný významný vztah neodhalily, což mohlo být způsobeno rušivými faktory, například vychýleným rozdělením dat nebo vlivem jiných mutačních biasů.

**Klíčová slova:** mutation bias, genetický kód, adaptivní evoluce, *Mycobacterium tuberculosis*

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# Glossary

- **Nucleotides (A, T, C, G, U)** – The monomeric units of nucleic acids:
  - A – Adenine
  - T – Thymine (DNA only)
  - C – Cytosine
  - G – Guanine
  - U – Uracil (RNA only, replaces thymine)
- **Adaptive mutation** – A genetic change that increases the fitness of an organism and can become fixed in a population through positive selection.
- $N_1 \leftrightarrow N_2$  – A type of point mutation in which one nucleotide  $N_1$  is substituted with another nucleotide  $N_2$  or vice versa. It is symmetrical and does not indicate directionality.
- $R_1:Y_1 \rightarrow R_2:Y_2$  – A directional substitution where one purine-pyrimidine base pair  $R_1:Y_1$  is replaced by a different purine-pyrimidine base pair  $R_2:Y_2$ . This notation reflects the final paired state after the mutation is fixed in the genome.
- **Transition** – A point mutation involving a purine-to-purine ( $A \leftrightarrow G$ ) or pyrimidine-to-pyrimidine ( $C \leftrightarrow T$ ) substitution.
- **Transversion** – A point mutation in which a purine is replaced by a pyrimidine, or vice versa (e.g.  $A \leftrightarrow C$ ,  $A \leftrightarrow T$ ,  $G \leftrightarrow C$ ,  $G \leftrightarrow T$ ).
- **Mutation rate** – The probability that a mutation occurs at a specific site per generation. It reflects the overall frequency of mutational events in the genome.
- **Mutation spectrum** – The relative frequencies of different types of mutations occurring in a given organism, genomic context, or experimental setting. It describes what kinds of mutations occur (e.g. transitions vs. transversions, insertions, deletions), rather than how often mutations occur overall.
- **Mutation supply** - The total number of new mutations entering a population per generation, typically calculated as the product of the population size and the per-genome mutation rate.
- **Mutation bias** – A non-random pattern in mutation occurrence, where some types of mutations arise more frequently than others due to biochemical, structural, or contextual factors.
- **Mutator strain** – A bacterial strain with an elevated mutation rate, typically due to defects in DNA repair mechanisms.
- **Mutation accumulation experiment** – A type of experimental design in which organisms are passed through many generations under minimal selection pressure, allowing spontaneous mutations to accumulate. This approach is used to estimate mutation rates and detect mutational spectra without a strong selection.

# Introduction

Evolution does not happen in one, but rather in two steps (Stoltzfus & Yampolsky, 2009). First, the origin of genetic variation through mutation, and second, either acceptance or rejection of this novelty through natural selection and genetic drift (Laland et al., 2015). While natural selection plays a central role in shaping which variants become fixed by choosing which option enhances the fitness the most, it only chooses among the presented set of genetic variants. The concept of mutation bias highlights that the set of genetic variants available for selection is not the result of a uniform or entirely random process. Rather, mutations tend to arise more frequently in certain directions than others, due to various biochemical and structural factors (Stoltzfus & Yampolsky, 2009).

Mutation biases have long been recognised. Transitions are more common than transversions, there is a widespread bias favouring G:C  $\rightarrow$  A:T changes, often associated with the high mutability of CpG dinucleotides, and small deletions occur more frequently than small insertions. Moreover, the sequence and structural context can influence local mutation spectrum. While additional sources of mutation bias remain to be clarified, it was traditionally believed that such biases were irrelevant to evolution, where selection was seen as the main driver. However, recent evidence suggests that mutation bias may shape evolutionary outcomes alongside selection.

If mutation bias influences adaptation, its recurring patterns may significantly affect how predictable evolution is. A thorough understanding of these biases is essential for interpreting the genetic differences that have accumulated among species and individuals (Keller et al., 2007) as well as evolutionary forecasting. Apart from evolutionary theory, insights into mutation bias are also highly relevant in applied contexts, such as improving our ability to anticipate the evolution of cancer, predicting resistance pathways in pathogens and pests, and guiding strategies to manage infectious diseases more effectively (Cano et al., 2023).

A key step in demonstrating the role of mutation bias in adaptive evolution is detecting such bias among adaptive mutations themselves. The goal of this thesis is therefore to present and examine several studies in which these patterns have been documented. In addition, we test a hypothesis that amino acid changes accessible through a greater number of pathways, as defined by the structure of the canonical genetic code, may be overrepresented among adaptive substitutions.

This thesis is structured as follows:

- The first chapter introduces the main types of mutation bias and their molecular origins. This is followed by a historical overview of how the evolutionary significance of mutation bias has been interpreted over time. The third section outlines the theoretical background and key factors that influence the strength of mutation bias, and finally, the fourth section reviews empirical studies that have identified mutation bias among adaptive mutations.
- The second chapter of the thesis investigates the potential role of the genetic code in shaping the mutation spectrum and assesses its influence on a dataset of adaptive mutations from *Mycobacterium tuberculosis*.

# Literature Review and Conceptual Background

## 1. Types and Molecular Causes of Mutation Bias

Mutation biases arise from various molecular, environmental, and structural factors that influence which mutations occur more frequently than others. The following section aims to provide a literature overview of these biases, along with the biochemical mechanisms that underlie them.

### 1.1. Single-nucleotide Substitution Rates

On the DNA level, there are 6 possible single-nucleotide substitutions,  $A \leftrightarrow G$ ,  $C \leftrightarrow T$ ,  $A \leftrightarrow T$ ,  $A \leftrightarrow C$ ,  $C \leftrightarrow G$ ,  $G \leftrightarrow T$ . One could expect all of them to occur with equal probability. However, some of them are observed more frequently than others, as visualised in Figure 1.

#### 1.1.1. Transition Bias

Each nucleotide has one possible transition and two possible transversions. When assuming equal mutation rates of both types of substitutions, the transition-to-transversion ratio should

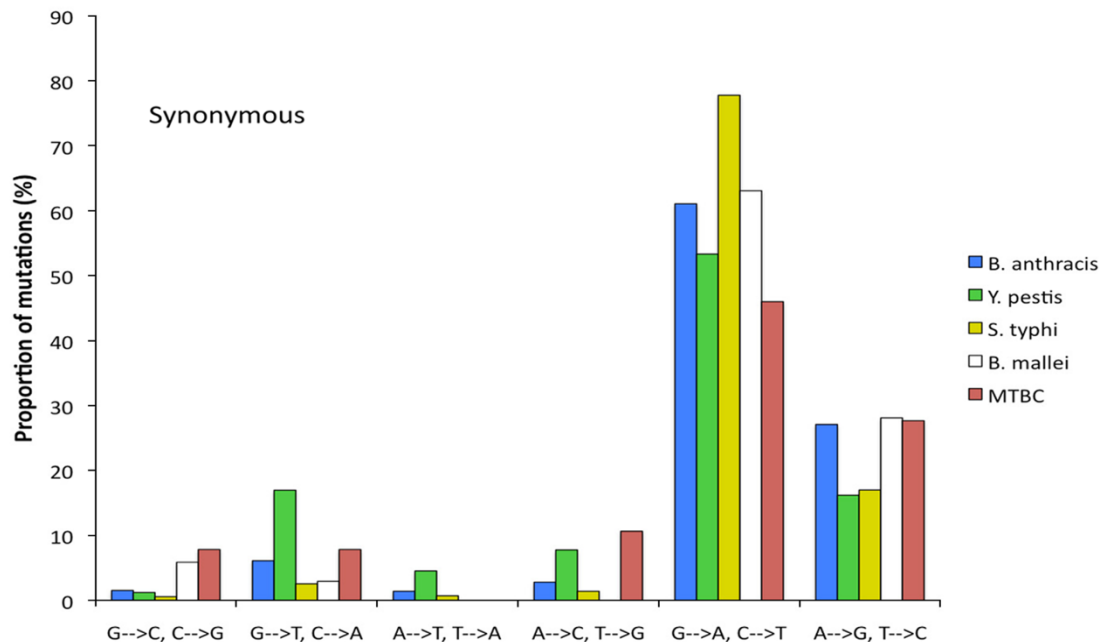


Figure 1. Relative rates of the six types of nucleotide pair substitutions within synonymous sites across species. The figure highlights that  $G:C \rightarrow A:T$  transitions are consistently the most common, even when selection is minimal, illustrating the underlying mutational bias. Taken from Hershberg & Petrov (2010), Fig. 2 (B).

be 1:2, i.e. 0.5. It may be more relevant to focus only on missense mutations. There are 116 possible transitions and 276 possible transversions that lead to amino acid exchange (Stoltzfus & Norris, 2016). That means that the transition missense mutation ratio should be approximately 0.42. Therefore, under a null hypothesis, we should observe this 0.5, respectively 0.42 ratio in DNA sequences. However, empirical studies have repeatedly shown a deviation from this null expectation.

Multiple studies have consistently reported transitions at approximately two to five times higher frequencies compared to transversions (Katju & Bergthorsson, 2018; Ossowski et al., 2010; Payne et al., 2019; Stoltzfus & Norris, 2016; Wakeley, 1996; Y. Zhu et al., 2014). Although the strength of transition bias may differ between organisms due to variations in DNA replication fidelity, repair mechanisms and mutagenic environments, it is observed almost universally across taxa, as can be seen in Figure 2 (Lynch et al., 2023). Exceptions are rare, such as in grasshopper pseudogenes (Keller et al., 2007).

As Lynch et al. (2023) point out, a wide range of factors can influence mutation spectrum. Among these, multiple DNA polymerases, DNA repair proteins, and genes whose products affect the mutagenicity of the intracellular environment play significant roles. Simply put, it is about which mutations are more likely to arise and more likely to escape repair mechanisms (Sane et al., 2023).

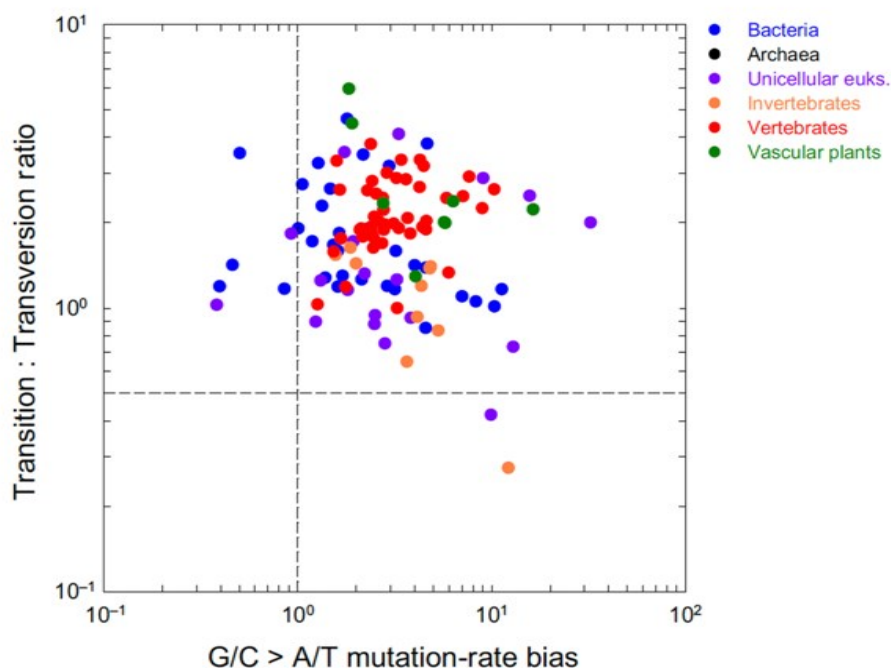


Figure 2. Distribution of transition-to-transversion ratios and their relation to  $G:C \rightarrow A:T$  mutation-rate bias across various domains of life, indicating that a bias favoring transitions is prevalent and phylogenetically widespread. Dashed lines indicate the null expectation assuming equal mutation rates for all substitution types, i.e. 0.5 and 1.0. Figure taken from Lynch et al. (2023), Fig. 4 (A).

A key mechanism leading to transitions, particularly C:G  $\rightarrow$  T:A substitutions, is the spontaneous deamination of cytosines (Rosenberg et al., 2003). It leads to uracil, which is however not naturally present in DNA and can be more easily recognised and excised by the enzyme DNA-uracil glycosidase. True mutational hotspots for transition bias are 5-methylcytosine residues. If a cytosine is methylated, the spontaneous deamination leads to thymine, which cannot be excised by the same repair system (Duncan & Miller, 1980). 5-methylcytosine is more prone to deamination under conditions of oxidative stress, such as in the presence of free oxygen radicals (Lynch et al., 2023). This contributes to the fact that certain regions of the genome are more susceptible to this reaction than others. The phenomenon is particularly pronounced in the mitochondrial genome, mtDNA, where oxidative stress is more frequent and therefore mtDNA has a notably higher rate of C:G  $\rightarrow$  T:A transitions than nuclear DNA (Katju & Bergthorsson, 2018; Lynch et al., 2023; Payne et al., 2019; Wakeley, 1996).

Furthermore, tautomeric shifts during DNA replication can more easily produce transitions due to the chemical similarity within purines or pyrimidines, making mispairing more likely (Stoltzfus & Norris, 2016). Transitions tend to cause less disruption to the local geometry of DNA base pairing compared to transversions, which more severely alter the shape of the double helix and are thus more likely to be recognised by DNA repair mechanisms (Guo et al., 2017; Lynch et al., 2023). Moreover, proofreading and mismatch repair enzymes are generally more effective at recognising and correcting transversions, further contributing to the elevated fixation probability of transitions (Lynch et al., 2023).

### 1.1.2. CpG Dinucleotide Hotspots and AT Bias

Apart from transitions being favoured, substitutions to adenosine and thymine are more frequent than to cytosine or guanine (Ossowski et al., 2010; Y. Zhu et al., 2014). This phenomenon is commonly referred to as AT mutation bias and is widely observed in various taxa, mainly in bacteria. For instance, Katju & Bergthorsson (2018) report that wild-type *Escherichia coli* shows a G:C  $\rightarrow$  A:T mutation rate that is 1.2-fold to 2-fold higher than the reverse, leading to an overall mutational pressure favouring AT enrichment. Similarly, Lynch et al. (2023) introduce the parameter  $\beta$ , which quantifies AT bias as the ratio of G:C  $\rightarrow$  A:T to A:T  $\rightarrow$  G:C mutation rates. In most organisms studied,  $\beta$  exceeds the neutral expectation of 1.0 and can reach values as high as 4.0, underscoring the prevalence of this bias (See Figure 2).

The AT mutation bias can also be attributed to the biochemical processes of mutation, particularly in GC-rich regions of the genome, specifically so-called CpG dinucleotides, which are common in Metazoans and are especially prone to cytosine methylation and subsequent C:G  $\rightarrow$  T:A mutation (Katju & Bergthorsson, 2018; Ossowski et al., 2010; Storz et al., 2019). Similarly, spontaneous deamination can also happen to 8-oxoguanine, which mispairs with adenine, resulting in G:C  $\rightarrow$  T:A transversion. Moreover, AT bias can also be attributed to

UV-induced mutagenesis, particularly at dipyrimidine sites, as reported by Ossowski et al. (2010) in *Arabidopsis thaliana*.

The effect of the CpG dinucleotide mutational hotspot is highly significant. For example, in mammals, CpG transitions occur at rates approximately twofold higher than non-CpG transitions, while CpG transversions are also elevated and occur at rates comparable to non-CpG transitions, indicating that CpG sites are broadly hypermutable across substitution types (Siepel & Haussler, 2004).

## **1.2. Indel Biases**

In addition to base substitutions, insertions and deletions (indels) represent a distinct class of mutations that exhibit biases and structural dependencies. Indel rates may disproportionately affect sequences depending on GC content (Katju & Bergthorsson, 2018). A deletion bias has been observed in most prokaryotes and may contribute to the maintenance of their compact genomes (Mira et al., 2001). Similar trends have been observed in eukaryotes, though modulated by factors such as genome size, chromatin structure, and the complexity of DNA repair mechanisms (Katju & Bergthorsson, 2018).

A broader perspective on the impact of indel biases is provided by Luiselli et al. (2025), who explored how insertion bias contributes to the equilibrium size of non-coding genome fractions. Specifically, they developed a mathematical model that predicts the non-coding content of a species' genome based on its specific mutation rates and effective population size. The model relies on two opposing forces: insertion and duplication bias that increases the genome size and the higher susceptibility of larger genomes to structural mutations, which reduce it.

## **1.3. Context-Dependent Mutation Bias**

The mutation spectrum can further be influenced by both the sequence and structural context. Substitution rates have been shown to depend not only on the identity of the mutated base itself, but also on the nucleotides that surround it. This context dependence is evident across both coding and non-coding regions and includes hotspot sites like CpG dinucleotides, as well as more complex and subtle dependencies (Horton et al., 2021; Siepel & Haussler, 2004; Sung et al., 2015). In addition to biochemical and sequence factors, the physical structure of nucleic acids can also create conditions that favour certain mutations over others. Numerous studies have reported that parallel evolution often targets the same genomic regions, frequently involving recurrent mutations within a narrow range of loci or even individual nucleotide sites (Horton et al., 2021; references therein: [7-12]).

Discussions are ongoing about other factors as well, however without definite conclusions. For example, there appears to be a phenomenon known as strand bias, meaning that the like-

likelihood of a certain single-nucleotide mutation depends on which DNA strand, leading or lagging, a given nucleotide is located on. The likelihood of different mutations occurring depends on the strand context, as each DNA strand is replicated by a distinct DNA polymerase with different fidelity, and repair mechanisms also vary in efficiency between strands (Lujan et al., 2014). For example, A:T → G:C transitions occur more frequently on the lagging strand template in *E. coli*, suggesting a structural or mechanistic cause associated with replication (Lee et al., 2012; Shewaramani et al., 2017). Moreover, in mismatch-repair-deficient strains of *Pseudomonas aeruginosa*, indels occurred more frequently when guanine was on the lagging strand template (Dettman et al., 2016).

Furthermore, several studies suggested a relationship between transcription rates and mutation rates. Some say that higher levels of transcription are associated with increased mutation rate (Hudson et al., 2003), others argue that elevated transcription may actually reduce mutation rates due to the action of transcription-coupled repair mechanisms. Therefore, conclusions regarding the relationship between transcription rate and mutation rate and spectrum remain uncertain (Katju and Bergthorsson, 2018 and references therein). Additionally, Monroe et al. (2022) provide evidence that mutation rates are lower across all functionally important genomic regions, including highly transcribed genes, suggesting that mutation rates may evolve under selection to protect essential parts of the genome.

## **2. Historical Perspective on Mutation Bias**

A wide range of factors are known to cause mutation bias, making it evident that the mutation spectrum is inherently unbalanced. However, the key question remains: what role does this play in evolution? Views on this matter have long varied, and the following section outlines how perspectives on mutation bias have developed over time.

### **2.1. Modern Synthesis**

Modern evolutionary biology would not be the same without the findings of two significant scientists, Charles Darwin and Gregor Johann Mendel. They both worked in the 18th century, Mendel defined the foundation principles of genetics, and Darwin came up with a revolutionary explanation of evolution by natural selection (Bowler, 2015). However, society had to wait almost another hundred years until someone connected these findings. This interconnection took place between the 1920s and 1950s and Julian Huxley (1942) called it the Modern Synthesis.

The main idea of this school of thought was that natural selection is not only compatible with Mendelian inheritance but acts directly upon genetic variation itself. It described natural selection as the main force, which shifts the frequencies of genes in an existing set of variants and therefore enables adaptive changes in populations over time (Dobzhansky, 1937; Fisher, 1930; Haldane, 1932; S. Wright, 1931). It was understood that the origin of genetic diversity

comes from mutation. However, mutation was believed to be unable to overcome the force of opposing selection, as mutation rates are relatively low in contrast to selection coefficients (Fisher, 1930, pp. 78 - 80).

In this framework, adaptation was thought to proceed particularly through combinations of alleles of small effects in various loci, generated and reshuffled by recombination. The only scenario in which mutation was regarded as a significant factor was within the framework of neutral evolution, when it was not opposed by natural selection (Haldane, 1927).

## **2.2. Biases in Introduction of Variance**

In the post-synthesis era of evolutionary biology, many aspects of the Modern Synthesis were either confirmed or questioned. Revolutionary discoveries in molecular biology during the second half of the 20th century revealed new possibilities for studying evolution. It became clear that molecular sequences of amino acids and nucleotides serve as records of evolutionary history (Mayer & Craig, 2013), leading to the rapid development of sequencing methods. Combined with advances in computational technology, this progress enabled the creation of bioinformatics tools such as sequence alignment, phylogenetic tree construction, and conserved pattern detection, which form the foundation of modern phylogenetic research (Higgs & Attwood, 2013).

New insights from molecular studies, genomics, ecology, and social science led to doubts about the completeness of the Modern Synthesis (Laland et al., 2014). Jablonka & Lamb (2007), in line with Laland et al. (2014, 2015), argued that natural selection can act not only on genetic variation but also on epigenetic, behavioural, and symbolic inheritance. A closely related perspective has emerged from the field of evolutionary developmental biology, which has also experienced significant development since the mid-20th century (Mayer & Craig, 2013). It further challenges the traditional view by proposing that developmental constraints can bias the variation available for selection, thereby acting as another force in shaping adaptive evolution (Laland et al., 2014; Stoltzfus & Yampolsky, 2009).

Similarly, another bias in the introduction of variation was suggested. Specifically, King et al. (1971) stated that mutations may not be random, on the contrary, that emergence of mutations can create a directional bias. That was in contrast to Modern Synthesis, where mutation was portrayed as a passive source of variation. King was not the first to come up with this idea, Hugo de Vries already suggested in his work “Die Mutationstheorie” (1901) that mutation might have an important role in determining the direction of evolution. However, he also claimed that speciation arises through large mutations. Therefore, his work was only partially correct and was later overshadowed by Modern Synthesis (Bowler, 1978). Similarly, King’s work was not very influential at first because he did not present enough empirical evidence to support his claims.

This situation changed, when Yampolsky & Stoltzfus (2001) provided mathematical models, which challenged the classical assumptions that mutations are purely random. Stoltzfus and colleagues continued to expand the evidence of mutation bias. In 2009, they published a study analysing how mutation shapes the variance in mean rates of amino acid replacement during human–chimpanzee divergence (Stoltzfus & Yampolsky, 2009).

Further studies in the 21st century confirmed the claims made by Stoltzfus and Yampolsky (Bailey et al., 2017; Laland et al., 2015; Lynch, 2007; Nei, 2013; Noble, 2017). However, it has not been generally accepted in the scientific community, and discussions about the importance of mutations in evolution are still ongoing. Svensson & Berger (2019) argue that while mutation bias interacts with selection, it cannot independently drive evolution. As they proclaim, mutation can lead the direction of evolution only when specific conditions are met, namely, small population size and low level of standing genetic variation. In general, they emphasise the dominant role of natural selection as the only directional evolutionary force, determining which variants become fixed and thus contribute to phenotypic change.

Overall, despite ongoing debates about the importance of mutation bias, it is now widely recognised as a relevant part of evolutionary theory. Rather than replacing the principles of the Modern Synthesis, it expands upon them, demonstrating that evolution is not only shaped by selection acting on random mutations but also by the biases in how mutations arise and spread.

### **3. Theory of Mutation Bias in Adaptation**

How could mutation bias influence adaptive evolution? The relationship between the two steps of evolution, the origin of genetic variation and its subsequent fate determined mostly by selection and genetic drift, is not universal. While the previous section traced the historical perspectives on this question, the following paragraphs will outline the specific factors that modify the evolutionary effects of mutation bias as understood today.

A key factor determining the influence of mutation bias on adaptation is the mutation supply, defined as the product of population size and mutation rate. Based on a combination of empirical data and mathematical models, Cano et al. (2022) report that the predictive power of mutation spectrum depends strongly on the number of beneficial mutations available in the population. When the supply of beneficial mutations is high, either due to large population size or elevated mutation rate, multiple adaptive variants can arise simultaneously and compete, a phenomenon known as clonal interference. This competition reduces the likelihood that mutations favoured by mutation rates will be fixed, thus weakening the impact of mutation bias. Furthermore, if sufficient polymorphism accumulates within a population, resulting in high standing genetic variation, the opportunity for the introduction of new mutations is substantially reduced, thereby reducing the impact of mutation bias. However,

such situations are relatively rare (Stoltzfus & Yampolsky, 2009). On the contrary, smaller populations experience reduced mutation supply. Under these conditions, beneficial mutations are rare, and evolution typically follows the so-called origin-fixation regime. The first mutation that arises, often the one favoured by mutation bias, is more likely to fix, even if it is not the most advantageous (Svensson & Berger, 2019; Yampolsky & Stoltzfus, 2001).

Moreover, the influence of mutation bias is also associated with the mutational target size. This refers to the number of potential distinct mutations that can generate a particular beneficial phenotypic outcome, which determines the likelihood of arising of adaptive mutations. The bigger the target size, the greater the chance that mutation bias will shape which substitutions contribute to adaptation (Cano et al., 2022).

Other important factors include the strength of natural selection, fitness landscape architecture, whether adaptation is short-term or long-term, heterogeneity in the mutation rates across the genome and the presence of epistasis (Stoltzfus, 2006; Stoltzfus & McCandlish, 2017; Yampolsky & Stoltzfus, 2001). These variables together determine whether biased mutational input can significantly shape evolutionary outcomes.

In conclusion, mutation bias is not a universally noticeable factor in evolution. However, under certain conditions, such as a smaller population size and sufficient opportunity for the emergence of new genotypes, it is a relevant component of adaptation. It can affect which genetic variants arise more frequently and therefore restrict some evolutionary pathways. That is particularly important because it enhances the predictability of evolution (Cano et al., 2023; Stern & Orgogozo, 2009).

## **4. Empirical Evidence of Mutation Bias in Adaptation**

Multiple sources of mutation bias have been identified, including biochemical, sequence, and structural factors. Theoretical models suggest these biases can influence adaptive evolution by shaping which mutations arise. But does this occur in practice? A key step in demonstrating the role of mutation bias in adaptive evolution is its observation among adaptive mutations themselves. This section therefore presents several empirical studies that provide clear examples of mutation bias operating among adaptive mutations, demonstrating that it can play a meaningful role in steering evolutionary trajectories.

### **4.1. Analyses of Natural and Experimental Data**

The first section focuses on studies that analysed observed data from both natural and experimental environments to identify patterns of mutation bias in adaptive evolution across diverse organisms and settings.

Recent work by Cano et al. (2022) provided empirical support for the role of mutation bias in shaping adaptation. By combining experimental evolution data from three organisms, *Saccharomyces cerevisiae*, *E. coli*, and *M. tuberculosis* and computational models, they showed that adaptive substitutions are predictable from the underlying mutation spectrum. Moreover, they demonstrated that the mutation spectrum of a given organism can be conversely inferred from the observed specific spectrum of adaptive substitutions, further highlighting the predictive link between mutation processes and evolutionary outcomes.

This broad confirmation of the role of mutation bias highlights the need to investigate individual mutational asymmetries and their contribution to adaptive evolution.

#### **4.1.1. Transition Bias**

As mentioned in section 1.1.1, transition bias is among the most thoroughly studied forms of mutation bias. To investigate whether this mutational asymmetry also shapes adaptive evolution, Stoltzfus & McCandlish (2017) conducted a meta-analysis on this matter. They compiled two datasets of adaptive amino acid changes that have arisen at least twice in different lineages, to ensure the beneficial nature of the observed mutations. One dataset was from a laboratory experiments, where 63 amino acid substitutions arose independently a total of 389 times, while the other included 55 replacements that occurred independently 231 times in natural populations. In order to ensure that the observed substitutions were truly adaptive, the authors relied on published studies that provided either genetic associations with phenotypic traits, such as toxin resistance, or direct experimental validation, typically through site-directed mutagenesis or selection experiments following random mutagenesis. They distinguished between mutational paths and events, where mutational path is each distinct substitution in nucleotide sequences, while mutational event is individual observation of such replacement in a specific lineage. Across both experimental and natural datasets of parallel adaptation, transitions were significantly overrepresented relative to transversions, with transition to transversion ratios reaching up to 3.6 for events and 2.2 for paths in experimental cases, and 1.3 for events and 1.0 for paths in natural cases. That substantially exceeded the null expectation of 0.5 and supported the role of mutation bias in shaping adaptive evolution.

Furthermore, Stoltzfus & McCandlish (2017) examined limiting their data to only those paths that were observed multiple times, from at least three times to more than eight times. Such stricter criteria were expected to reduce the presence of hitchhiker mutations and other non-adaptive contaminants. Even then, within these more narrowly defined subsets, the findings remained qualitatively consistent and highly significant. Therefore, parallel adaptation appears to follow substitution paths that are favoured by underlying mutation biases.

A similar conclusion was reached by Payne et al. (2019). Even though Stoltzfus & McCandlish (2017) included both experimental and natural datasets in their analysis, the experimental data were limited to bacteria and their phages, whereas the natural data were drawn

only from multicellular eukaryotes. In particular, they did not cover whether transition bias plays a role in populations of bacterial pathogens evolving in nature, such as those evolving resistance to antibiotics. The study by Payne et al. (2019) directly addresses this deficiency by examining transition bias within adaptive substitutions leading to antibiotic resistance in natural populations of *M. tuberculosis*. That is a clinically relevant human pathogen and a leading cause of death from infectious diseases throughout history, which evolves its resistance to antibiotics exclusively through chromosomal mutations (Gagneux, 2018). Additionally, while Stoltzfus & McCandlish (2017) observed 389 and 231 events in experimental and natural datasets respectively, Payne et al. (2019) analysed two datasets with 2775 and 2671 events.

In line with previous findings, Payne et al. (2019) observed a significant deviation from the 0.5 null expectation of transition to transversion ratio for both mutational paths and events. Their observed transition to transversion ratio was over 3.4-fold higher at the event level and more than 1.4-fold higher at the path level compared to what would be expected under a neutral model assuming equal probabilities. Significantly, transition bias was also detected in gene promoters and in the 16S ribosomal RNA gene *rrs*, both of which do not encode proteins, as well as in synonymous mutations whose outcomes are functionally equivalent to the original state. Since natural selection usually plays little to no role in such cases, these observations imply that the consistent overrepresentation of transitions in these contexts is driven primarily by mutation bias. Besides that, they found that transition bias was consistently present across resistance mutations to nearly all tested antibiotics. As a result, the authors addressed the critical challenge of uncovering the mutational paths by which a major human pathogen adapts to antibiotics, a significant issue for both evolutionary biology and public health.

Several studies have attributed the prevalence of transitions to natural selection rather than to elevated mutation rates. From this perspective, transitions are considered selectively advantageous because they tend to preserve the properties of the original nucleotide and maintain the local base-pairing geometry of DNA (Guo et al., 2017, see also Section 1.1.1). Similarly, non-synonymous transversions tend to disrupt the biochemical properties of the original and mutant amino acids (Rosenberg et al., 2003) which also lead to more significant changes in the biochemical properties and functionality of the protein (Keller et al., 2007; Wakeley, 1996). These arguments collectively suggest that the overrepresentation of transitions among adaptive substitutions may reflect their lower disruptive effect on molecular structure rather than how frequently they occur. However, empirical evidence points more strongly to a primary role of mutation bias.

Clear evidence for the role of mutation bias is the prevalence of transitions among synonymous and non-coding substitutions, where selection is minimal (Payne et al., 2019). Furthermore, Stoltzfus & Norris (2016) conducted a systematic review of transition bias and

focused on the underlying reasons. They assembled 1239 mutants with a substitution leading to a change of amino acid, 544 of which were transitions and 695 transversions. Altogether, it covered 145 of the 150 observable missense single-nucleotide substitutions. They collected data from studies which measured fitness values of these mutants. They analysed the effects by using 3 individual predictors, which proved to be objective for effects of substitutions across loci and proteins. They observed that transitions are not significantly more fit or conservative than transversions. On the contrary, they mentioned four studies of beneficial mutations, which indicated lower fitness values for transitions compared to transversions. Overall, the probability that a transition mutation confers higher fitness than a transversion is 53% (95% CI: 50–56), which is only slightly above random expectation. This effect size is modest relative to most biochemical predictors and insufficient to account for the commonly observed several-fold excess of transitions.

In line with these findings, Lyons & Lauring (2017) suggest that transitions and transversions may exhibit similar fitness effects among neutral or mildly beneficial mutations. However, they argue that transversions tend to be significantly more harmful, and therefore more frequently eliminated by selection. Furthermore, it has been suggested that although transitions may not be generally more conservative across the full distribution of fitness effects, they could be overrepresented specifically among rare beneficial mutations, those in the extreme right tail of the fitness distribution (Stoltzfus & McCandlish, 2017).

To conclude, there is no consensus in current research as to whether the observed transition bias arises primarily from selection or mutation bias. As described by Stoltzfus & Yampolsky (2009), these two forces can be represented as vectors pointing in different directions, each contributing to the evolutionary outcome. Their relative influence ultimately depends on the specific context (Payne et al., 2019).

#### **4.1.2. CpG Hotspot Sites**

As previously mentioned, some genome regions are more prone to mutation than others, and CpG dinucleotides are known mutation hotspots. Genetic studies of high-altitude birds found several adaptive changes affecting haemoglobin function to occur at CpG sites. This observation raises the possibility that CpG-related mutation bias may influence the evolution of increased affinity of haemoglobin to oxygen in response to hypoxia (Galen et al., 2015; X. Zhu et al., 2018).

Building on these observations, Storz et al. (2019) focused on the CpG dinucleotide regions in avian species and carried out a comprehensive study that directly tested the hypothesis that mutations at CpG dinucleotides contribute disproportionately to adaptive changes in haemoglobin function. Using a paired-lineage design, they observed that high-altitude taxa exhibited significantly higher haemoglobin–oxygen affinity than their lowland relatives in 35

phylogenetically independent pairs of related bird species. Each pair that demonstrated a significant difference in haemoglobin affinity was subjected to closer examination. Specifically, they used reconstructed ancestral sequences and site-directed mutagenesis to identify the specific amino acid substitutions responsible for increased oxygen affinity in highland taxa. They observed 10 different mutational paths leading to a higher haemoglobin affinity with 1 to 6 events along each path, together 22 replacements that enhanced affinity. Out of these, 10 events and 6 paths involved a CpG substitution. That was significantly above the predictions of a null model, which assumed that the CpG status of a mutation is irrelevant to its likelihood of contributing to the adaptation. The null model predicted only approximately 1 CpG path and 2 CpG events. Given that it is unlikely the mutations at CpG sites conferred unusually high fitness advantages compared to other possible substitutions with similar functional effects, the results of this study support the view that mutation bias, specifically, the elevated mutation rate at CpG dinucleotides, can influence the trajectory of adaptation.

## **4.2. Experimental Manipulation of Mutation Spectra**

Although statistical analyses of natural and experimental data are highly informative, they often cannot distinguish correlation from causation. The strongest evidence for the causal role of mutation bias in shaping adaptive evolution comes from experimental studies that deliberately manipulate the mutation spectrum, enabling researchers to observe how such changes influence evolutionary trajectories under controlled conditions (Stoltzfus & McCandlish, 2017).

### **4.2.1. DNA Sequence and Structural Context**

CpG dinucleotides are not the only sites prone to becoming mutation hotspots. Parallel evolution has frequently been observed to target the same genomic regions, often involving recurrent mutations at specific loci or even individual nucleotide sites (Horton et al., 2021; references therein: [7-12]).

Horton et al. (2021) focused on clarifying this phenomenon by analysing the origins of such hotspots. They studied two strains of soil bacteria *Pseudomonas fluorescens*, AR2 and Pf0-2x, which share a homologous genetic background and were engineered to be unable to use a flagellum. However, under strong directional selection, both of them were able to quickly evolve motility using the flagellum (Taylor et al., 2015). AR2 achieved the adaptive phenotype repeatedly via de novo hotspot mutation in the *ntrB* locus of the nitrogen regulatory (*ntr*) pathway, with over 95% of the replicate lines acquiring the exact same nucleotide mutation in the *ntrB* gene, A289C. Meanwhile, Pf0-2x evolved the motility via mutations in distinct sites of the *ntr* pathway.

Horton et al. (2021) conducted a series of experiments to investigate why the second strain does not evolve through the mutation hotspot, despite being subject to the same selective

pressure. They observed that the level of parallelism could not be explained by low accessibility of other mutational paths in AR2. On the contrary, they discovered that there are multiple pathways to achieve flagellum-motility but the A289C substitution significantly prevails even under different conditions. Furthermore, they compared the fitness of different adaptive variants by letting them compete in various environments but found that the *ntrB* A289C mutation offered no significant advantage. When introduced later than other evolved lines, it was even outcompeted, suggesting that its prevalence is at least partly due to factors beyond selection.

Subsequently, they traced the difference between the two strains to six synonymous substitutions in the DNA sequences surrounding the hotspot locus. Silent changes can affect local mutation rates either by altering the nucleotide triplet to one with a higher mutation rate or by modifying the DNA secondary structure. For example, they can change the curvature of the DNA or induce the formation of single-stranded DNA hairpin structures near reverse-complementary sequences (De Boer & Ripley, 1984; Duan et al., 2018). In such cases, the nucleotides that remain unpaired after the formation of a hairpin have been observed to exhibit increased mutation rates (B. E. Wright et al., 2003). The six observed silent mutations likely altered hairpin dynamics, reducing the mutational probability at this hotspot in Pf0-2x. By introducing the six synonymous substitutions from Pf0-2x into AR2, Horton et al. (2021) eliminated the hotspot effect entirely. However, the modified lines still adapted, only via different mutational paths. Conversely, inserting AR2's synonymous sequence into Pf0-2x induced parallelism in the *ntrB* hotspot, from 0% to 80%. These findings demonstrate that local synonymous sequence context, without altering the protein, can strongly bias which mutations occur, revealing mutation bias as a key driver of highly repeatable evolutionary outcomes.

#### **4.2.2. Mutator Strains**

While previous studies have shown that local sequence context can bias mutation patterns, another approach is the use of so-called mutator strains. These strains, characterised by exceptionally high mutation rates, pose a significant challenge in clinical infections, as they can rapidly evolve further adaptations. However, each mutator strain tends to disproportionately elevate specific types of mutations, such as transitions, transversions, or frameshifts, due to defects in particular pathways which normally prevent such errors (Couce et al., 2015; Sane et al., 2023). These mutation biases provide a valuable opportunity to experimentally assess how mutation supply and spectrum shape the outcomes of parallel adaptation. One such study was conducted by Couce et al. (2015), who examined how different mutational spectra shape evolutionary trajectories in *E. coli* under selective pressure. They examined three *E. coli* strains: a wild-type and two mutator strains. They hypothesised that mutational specificity might make mutators especially sensitive to genetic constraints.

The authors tracked the evolution of resistance to  $\beta$ -lactam antibiotic cefotaxime (CTX) in the three *E. coli* strains by monitoring changes in the well-studied  $\beta$ -lactamase gene *bla*<sub>TEM-1</sub>. This gene is known for its natural diversification through point mutations into numerous variants with enhanced activity against extended-spectrum  $\beta$ -lactam antibiotics such as CTX. In clinical isolates, CTX resistance mutations are predominantly G:C  $\rightarrow$  A:T transitions, precisely the type of mutations favoured by one of the mutator strains, *mutH*. The second one was *mutT*, which in contrast, elevates A:T  $\rightarrow$  G:C transversions, none of which is common within CTX resistance mutations found in natural isolates.

Couce et al. (2015) propagated each strain in two conditions, differing in the number of *bla*<sub>TEM-1</sub> copies, either 1 or 18, with progressively increasing concentrations of CTX. By including settings with 18 gene replicates, they increased the mutational target size and therefore the probability of rare beneficial mutations occurring. They expected the performance of *mutT* to depend on its ability to explore rare mutational paths compatible with its narrow mutation spectrum. If resistance primarily arose through common transition mutations, *mutH* would be expected to outperform the others, with wild-type strains adapting occasionally, and *mutT* adapting rarely due to its transversion bias.

However, the results revealed a different outcome. All three strains achieved resistance, with both mutator strains performing similarly and *mutT* even exceeding the others in the number of resistant populations. The presence of multiple plasmidic copies of *bla*<sub>TEM-1</sub> notably enhanced the adaptive potential of the wild-type strain, increasing the likelihood of rare beneficial mutations occurring, as the authors had anticipated. However, the mutator strains performed well regardless of copy number, owing to their higher mutation rates. Resistance arose not only from mutations in *bla*<sub>TEM-1</sub> gene but also from changes in other loci, notably in the *ftsI* gene, which provided a broad mutational target. All resistant variants carried a substitution in the sequence of at least one of these two genes.

As shown in Figure 3, each strain followed distinct adaptive trajectories that reflected their specific mutational spectra. The authors highlight mutation bias as a key deterministic force in evolution, guiding adaptation along the most mutationally accessible pathways. They emphasise the importance of mutational target size and mutation rates in shaping these outcomes and suggest that genetic constraints may underlie the high repeatability and predictability often observed in microbial evolution. This has practical implications, particularly in clinical settings, where mutator strains are already recognised as key drivers of antibiotic resistance due to their elevated mutation rates. Their spectrum-driven constraints may further complicate efforts to anticipate or control resistance development.

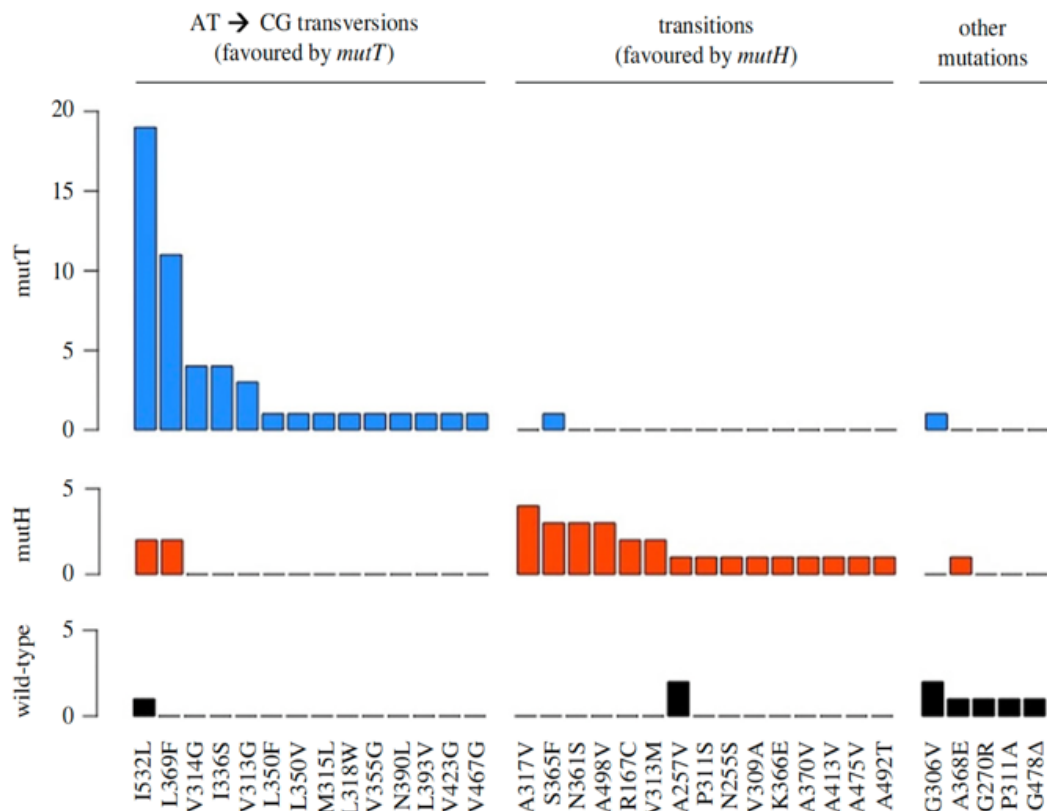


Figure 3. Variety of substitutions acquired by cefotaxime-resistant variants of *E. coli* from three different strains. *mutT* (blue) mutator strain elevates the rates of  $A:T \rightarrow G:C$  transversions, while *mutH* (red) elevates transitions. And wild-type strain does not elevate any particular type of mutations. It is visible how the mutation spectrum shapes the adaptive spectrum. Figure from Cano et al. (2023), based on Couce et al. (2015)

#### 4.2.3. Shifts in Mutation Spectrum

Previous sections have shown that no single mutation spectrum can be considered universally advantageous. In particular, as was discussed in the context of transition bias, transitions are not consistently favoured by selection, they simply occur more frequently. Moreover, mutator strains are capable of adapting regardless of their mutation spectrum, typically by following different evolutionary paths, which suggests that the spectrum is not inherently limiting (Couce et al., 2015). However, what has emerged as potentially beneficial is shifting the mutation spectrum itself (Sane et al., 2023).

Sane et al. (2023) explored this idea by examining how altered spectra affect adaptation in mutator strains. They demonstrated how a loss-of-function mutation in a regulatory gene involved in DNA repair can reshape the entire mutation spectrum of the organism. They applied this knowledge in their experiments, where they targeted deletions in DNA repair genes of *E. coli* strains. In particular, they employed *mutY* strain, which alters the spectrum without substantially increasing the overall mutation rate. This experimental setup allowed

them to isolate the effects of spectrum shifts from those of mutation rate alone. Consequently, they conducted mutation accumulation experiments followed by whole-genome sequencing to characterise the distribution of fitness effects in both wild-type and mutator backgrounds. Additionally, they tested the pleiotropic effects of individual mutations across multiple environments and supported their experimental findings with evolutionary simulations and phylogenetic analysis of DNA repair gene variation across bacterial species.

Their results show that modifying the mutation spectrum can be evolutionarily advantageous, as it enables organisms to explore mutations that were previously rare or inaccessible, thereby opening new paths for adaptation. Specifically, the *mutY* mutator strain generated a higher proportion of beneficial mutations compared to the wild type, despite a similar overall mutation rate. This advantage arose not from accessing more mutations, but from accessing different mutations. Moreover, phylogenetic analyses and simulation data suggest that shifts in mutation spectra may be more common in natural bacterial populations than previously assumed, often driven by horizontal gene transfer and recombination. This implies that shifts in base composition, such as increased AT content, may result from underlying mutation bias alone, rather than from direct selective pressure. In that case, the observed nucleotide composition of a species could largely reflect its mutational history rather than adaptive optimisation. These findings have further implications for the evolution of mutator strains, where changes in the mutation spectrum often accompany altered mutation rates. The evolutionary success of mutators depends among other factors on the supply of beneficial mutations, which can be strongly affected by changes in the spectrum, particularly in well-adapted populations.

Finally, the authors predict that shifting the mutation spectrum can have far-reaching evolutionary consequences. These may include faster access to beneficial mutations, a lower risk of mutational meltdown, and alternative adaptive trajectories, hypotheses that call for future theoretical and experimental testing.

In summary, the reviewed studies provide clear evidence that mutation bias can significantly shape the spectrum of adaptive mutations. This highlights its relevance for understanding the predictability of evolution and has important implications for applications such as forecasting antibiotic resistance or guiding experimental evolution.

# The Genetic Code as a Source of Mutation Bias

## 1. Hypothesis

As outlined in the previous chapter, mutation bias can arise for various reasons. From biochemical factors or DNA repair mechanisms that can cause bias in single-nucleotide mutations to structural reasons such as formation of hairpins, several layers of influence shape which mutations are more likely to occur. Since the genetic code plays a central role in translating genetic information into proteins, we suggest that its structure also leads to mutation bias. With a few exceptions, the canonical genetic code is a universally conserved mechanism used by cells to determine the translation of their genomes into proteins (Knight et al., 2001). It consists of 64 codons, each formed by 3 nucleotides. Each codon encodes either one of 20 amino acids or a stop signal. The numbers clearly show that some amino acids, such as leucine, serine, or arginine, are encoded by several codons, as can be seen in Figure 4. In contrast, methionine and tryptophan only have one codon each. Moreover, while single-nucleotide mutations in some codons can easily result in the exchange of amino acids, in other codons, they more frequently lead to silent mutations.

For example, methionine is encoded by only one codon, AUG. Therefore, each single-nucleotide mutation in the AUG codon leads to a change in amino acid. If a mutation appears in the third position, the change from guanine to each of the other nucleotides will result in methionine changing to isoleucine. However, if the uracil in the second position gets mutated, the ribosome will incorporate either arginine, lysine, or threonine. In case the adenine from the first spot is replaced, it can either lead to GUG and to valine or to leucine in the case of both UUG and CUG. To sum up, there are nine possible single-nucleotide mutations in codon AUG. Three of them result in methionine being replaced by isoleucine, two of them change it to leucine, and the four remaining lead each one to a different amino acid: lysine, threonine, arginine, and valine (See Figure 5).

An uneven distribution of potential mutational paths is present in 31 codons, which constitutes almost half of the 64 codons (See Figure 4). It is clear that the standard genetic code makes some mutational outcomes more accessible than others. This thesis explores whether such asymmetries are also reflected in the patterns of adaptive substitutions; specifically, whether the number of observed adaptive mutations somehow depends on the number of paths through which a given result can be obtained. In the following sections, the potential mutational paths from a codon to an amino acid, as determined by the structure of the genetic code, will be referred to as “genetic code pathways” for clarity.

|            |   | SECOND BASE |                              |     |           |     |               |     |            |   |            |
|------------|---|-------------|------------------------------|-----|-----------|-----|---------------|-----|------------|---|------------|
|            |   | U           |                              | C   |           | A   |               | G   |            |   |            |
|            |   |             |                              |     |           |     |               |     |            |   |            |
| FIRST BASE | U | UUU         | Phenylalanine                | UCU | Serine    | UAU | Tyrosine      | UGU | Cysteine   | U | THIRD BASE |
|            |   | UUC         | Phenylalanine                | UCC | Serine    | UAC | Tyrosine      | UGC | Cysteine   | C |            |
|            |   | UUA         | Leucine                      | UCA | Serine    | UAA | Stop Codon    | UGA | Stop Codon | A |            |
|            |   | UUG         | Leucine                      | UCG | Serine    | UAG | Stop Codon    | UGG | Tryptophan | G |            |
|            | C | CUU         | Leucine                      | CCU | Proline   | CAU | Histidine     | CGU | Arginine   | U |            |
|            |   | CUC         | Leucine                      | CCC | Proline   | CAC | Histidine     | CGC | Arginine   | C |            |
|            |   | CUA         | Leucine                      | CCA | Proline   | CAA | Glutamine     | CGA | Arginine   | A |            |
|            |   | CUG         | Leucine                      | CCG | Proline   | CAG | Glutamine     | CGG | Arginine   | G |            |
|            | A | AUU         | Isoleucine                   | ACU | Threonine | AAU | Asparagine    | AGU | Serine     | U |            |
|            |   | AUC         | Isoleucine                   | ACC | Threonine | AAC | Asparagine    | AGC | Serine     | C |            |
|            |   | AUA         | Isoleucine                   | ACA | Threonine | AAA | Lysine        | AGA | Arginine   | A |            |
|            |   | AUG         | Initiation Codon; Methionine | ACG | Threonine | AAG | Lysine        | AGG | Arginine   | G |            |
|            | G | GUU         | Valine                       | GCU | Alanine   | GAU | Aspartic Acid | GGU | Glycine    | U |            |
|            |   | GUC         | Valine                       | GCC | Alanine   | GAC | Aspartic Acid | GGC | Glycine    | C |            |
|            |   | GUA         | Valine                       | GCA | Alanine   | GAA | Glutamic Acid | GGA | Glycine    | A |            |
|            |   | GUG         | Valine                       | GCG | Alanine   | GAG | Glutamic Acid | GGG | Glycine    | G |            |

Figure 4. The canonical genetic code, modified from original Amino Acid Translation Table from Oregon State University. Highlighted in light orange are the codons, which have an uneven distribution of potential mutational paths. Together, there is 31 of them. With 3 stop codons in light blue, the highlighted codons make more than a half of all the sense codons.

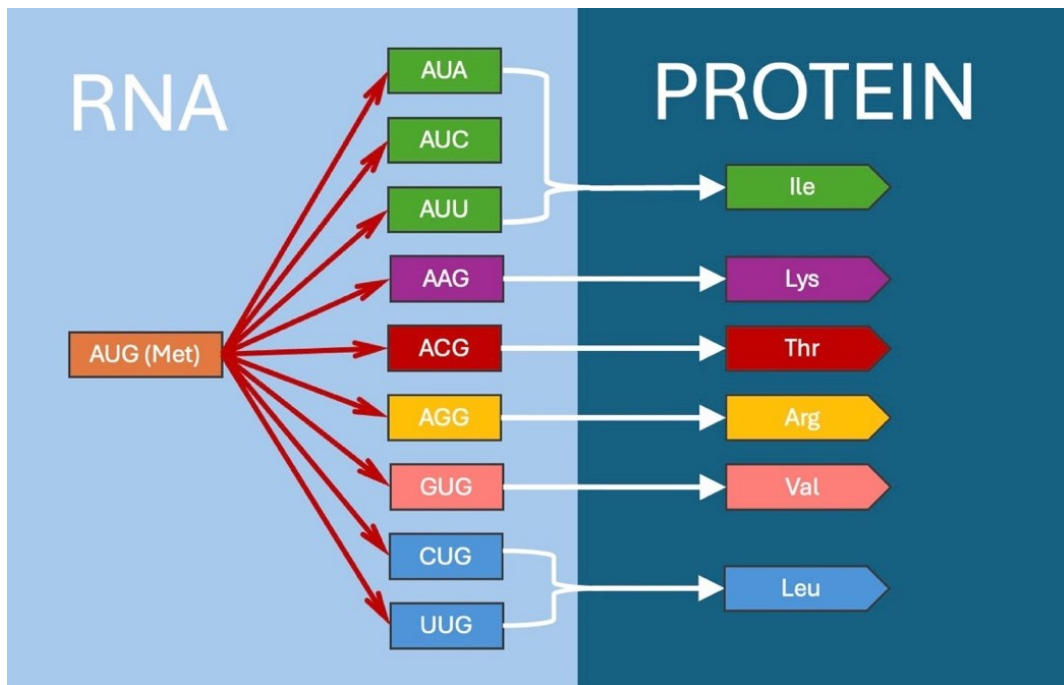


Figure 5. Example of all possible single-nucleotide mutations from codon AUG (methionine) and their corresponding amino acid translations. Each red arrow represents a mutation in the codon, leading to a different codon while each white arrow symbolises a translation of the new codon to an amino acid. Own illustration.

## 2. Existing Research

Stoltzfus & Yampolsky (2009) in their article about the importance of mutation bias mentioned several different sources of bias. Among others, they also considered the genetic code, specifically different number of possible genetic code paths from phenylalanine to leucine and tyrosine, illustrated in Figure 6. Moreover, they highlighted the following example. A serine  $\rightarrow$  proline substitution typically requires a T:A  $\rightarrow$  C:G transition, which is more frequent than the T:A  $\rightarrow$  A:T or G:C  $\rightarrow$  C:G transversions required for a serine  $\rightarrow$  threonine change, making amino acid change from serine to proline more accessible purely due to coding structure.

King, Le Cam, et al. (1971) also suggested that genetic code itself may contribute to mutation bias. Specifically, they pointed out that because amino acids differ in the number of codons that encode them, this unequal distribution inherently biases possible mutation outcomes. They demonstrated this by calculating nucleotide frequencies in mRNA sequences derived from vertebrate proteins. Their results showed that the structure of the genetic code inherently favours certain nucleotide compositions, thus potentially biasing the availability of specific mutations.

While the idea that the genetic code may contribute to mutation bias has been noted in the literature, no study has yet been found that would directly analyse this specific aspect among adaptive mutations. This thesis therefore aims to provide an initial insight into this question,

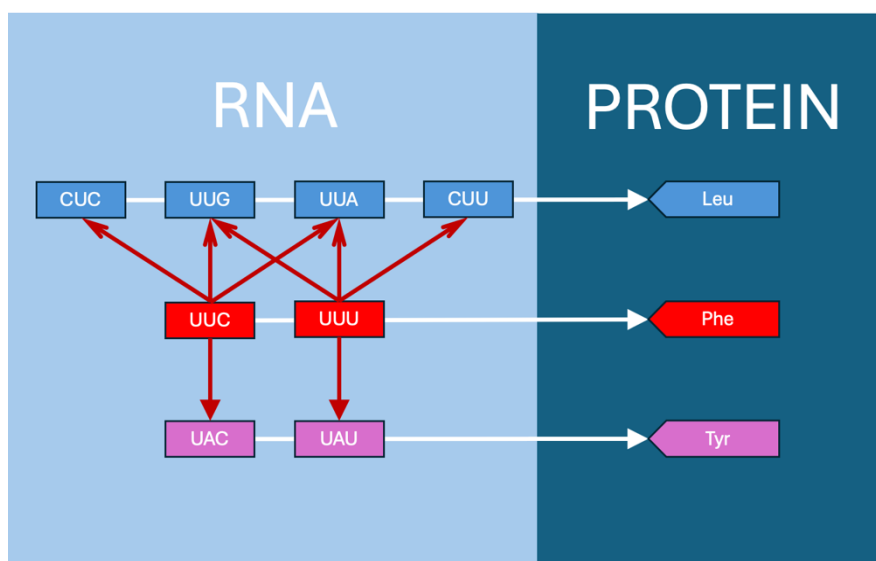


Figure 6. Illustration of single-nucleotide mutational paths connecting codons for phenylalanine (Phe) to leucine (Leu) and tyrosine (Tyr). Each red arrow represents a mutation in the codon, while each white arrow symbolises a translation of the new codon to an amino acid. The diagram shows that the number of available genetic code pathways differs depending on the target amino acid, reflecting potential mutation bias arising from the structure of the genetic code. Original figure created by the author, inspired by (Stoltzfus and Yampolsky, 2009, Fig. 7)

to test whether such a bias is detectable among adaptive mutations in *M. tuberculosis*, and to contribute to a broader understanding of how the structure of the genetic code may influence the spectrum of mutations observed in molecular evolution.

### 3. Materials

#### 3.1. Dataset and Preprocessing

The influence of the composition of the canonical genetic code on the ratio of observed mutations was tested on a dataset of single-nucleotide mutations known to confer antibiotics resistance observed in *M. tuberculosis*. We used the dataset collected by Payne et al. (2019), who originally analysed it in their study of transition bias in *M. tuberculosis*. It consists of originally two collections, one of them was compiled directly by Payne et al. (2019) and the other one was assembled by Manson et al. (2017). They gathered sequences with mutations, which were either experimentally confirmed to confer antibiotic resistance or have been strongly supported by previous literature. Moreover, they linked the variants to publicly available *M. tuberculosis* genomes.

Both datasets distinguished between mutational paths and mutational events. As mentioned before, a mutational path is each distinct change in nucleotide sequences, whereas a mutational event is an individual instance of such change along the path. Therefore, it is common to observe multiple events for one path in parallel adaptation. Payne et al. (2019) constructed a rooted phylogenetic tree from single-nucleotide polymorphisms in *M. tuberculosis* genomes. They focused on genomic positions associated with antibiotic resistance, assembled them into a multiple-sequence alignment, and combined this with the phylogeny to infer mutational gains and losses along evolutionary branches, as well as the frequency of events along each mutational path. Similar approach was employed by Manson et al. (2017).

Together, the dataset we used contains 4413 single-nucleotide mutational events. In particular, a single substitution occurs multiple times in the dataset if it emerged independently at several points in the phylogeny. All the events are associated with resistance to one of 11 antibiotics; therefore, they are considered adaptive and very unlikely to be contaminated by hitchhiker mutations or other non-causal variants.

We organised the list by codons and expanded it to include all possible single-nucleotide substitutions capable of producing an observable amino acid exchange. Therefore, synonymous substitutions and mutations from or to one of the stop codons were intentionally excluded, as these are unobservable. This gave us a total of 354 codon-to-amino-acid changes. For substitutions where empirical observations were not available, we assigned a value of zero. Furthermore, we extended the dataset to include the number of genetic code pathways through which each observed amino acid change could be achieved. For example, considering the previously mentioned AUG codon encoding methionine, three genetic code pathways

lead to a change from methionine to isoleucine - via three different codons, two pathways for the replacement of methionine by leucine, and only one path each for the mutation to lysine, threonine, arginine, and valine. The analysed dataset and code used for analysis and graph plotting is available online in [GitHub \(https://github.com/jaskovab/mutation-bias-thesis\)](https://github.com/jaskovab/mutation-bias-thesis).

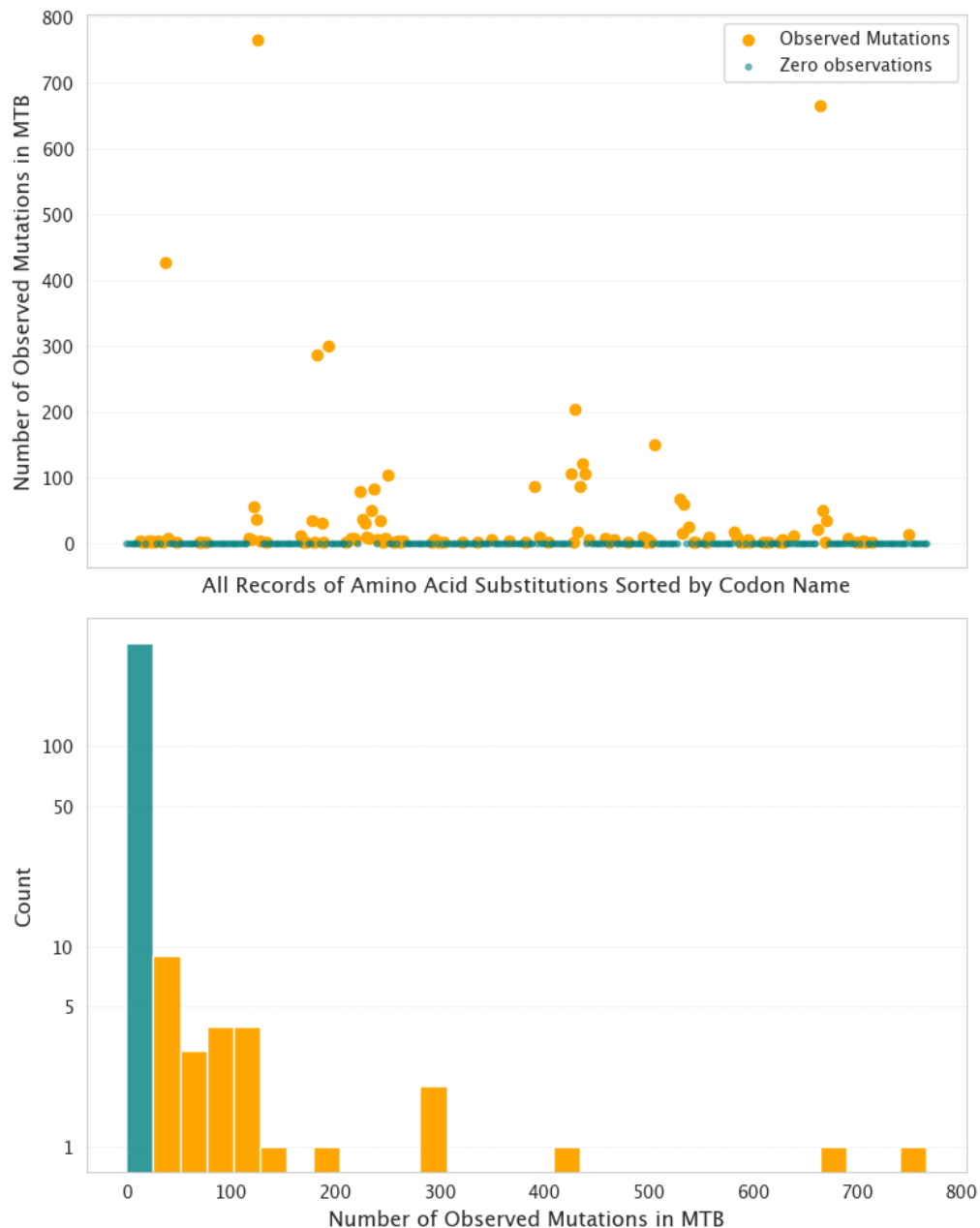


Figure 7. Distribution of mutation records. The upper panel shows the observed number of mutations in *M. tuberculosis* (MTB) for every possible single-nucleotide mutation that results in an amino acid substitution. The graph highlights how a few hotspot mutations disrupt the overall distribution of the data. The lower panel displays the frequency distribution of mutation counts (on a logarithmic scale). Most substitutions were never observed (highlighted in teal), while a small number occurred repeatedly (in yellow). The distribution is highly skewed, with a long tail caused by a few frequent mutation hotspots.

The dataset was strongly influenced by a small number of highly beneficial mutations, known as hotspot mutations (Horton et al., 2021). Of the total of 4413 observations, the 11 most frequent ones, which were all observed more than 100 times, account for 73.170% of all the records. Some changes were observed in low counts, while mutations that were not observed even once accounted for 68.644% of all possible changes (Figure 7).

### 3.2. Theoretical Framework - Modelling the Relationship

Phenotype, the outcome of evolution, is influenced by multiple interacting factors. Direct conclusions about the relationship between observed mutation counts and one specific element, such as the number of genetic code pathways, cannot be reliably drawn. Therefore, it is necessary to acknowledge other complexities and at least partially try to account for them. An attempt of expressing this relationship in a simplified, yet somewhat more realistic manner is described in the equation below.

$$\mathbb{E}[n(c, a)] \propto f(c)^{\beta_0} * \rho(c, a)^{\beta_1} * p_{acc}(c, a)^{\beta_2} \quad (1)$$

Where:

- $\mathbb{E}[n(c, a)]$  is the expected number of observed mutations that start with codon  $c$  and result in mutated amino acid  $a$
- $f(c)$  is the frequency of codon  $c$  in the genome of *M. tuberculosis*
- $\rho(c, a)$  is the number of possible single-nucleotide substitutions that lead from codon  $c$  to amino acid  $a$  according to the canonical genetic code
- $p_{acc}(c, a)$  is the probability of acceptance of this mutation in evolution
- $\beta_0, \beta_1, \beta_2$  are expressing the power of the dependance of the number of mutations on the individual factors

Frequencies of individual codons in the genome of *M. tuberculosis* are available in open sources, we used frequencies from strain CDC1551 which were compiled from GenBank sequences (Codon Usage Database: <https://www.kazusa.or.jp/codon/cgi-bin/show-codon.cgi?species=83331>, accessed 12. 3. 2025). It can reasonably be assumed that the frequency of mutations in a given codon is directly proportional to its frequency in the genome (Cano et al., 2022). Therefore, we assumed that the parameter  $\beta_0$  equals 1.

The probability of a mutation being accepted during evolution depends on natural selection and multiple external factors, including environmental conditions. Since these factors are too complex to quantify, it must be assumed that the acceptance probability is constant across all mutations, allowing this factor to be omitted from the equation.

Following this simplification, the relationship can be described as follows:

$$\mathbb{E}[n(c, a)] \propto f(c)^1 * \rho(c, a)^{\beta_1} \quad (2)$$

or

$$\frac{\mathbb{E}[n(c, a)]}{f(c)} \propto \rho(c, a)^{\beta_I} \quad (3)$$

### 3.3. Distributional Properties and Model Selection

In order to analyse the data, it is crucial to understand their distribution. Observed mutations could follow a binomial distribution because mutations at each nucleotide site represent discrete events with two possible outcomes - mutation occurring or not occurring. These events are independent of each other, have a consistent probability of occurrence at each site, and the total number of sites remains constant, thus meeting the key assumptions necessary for the binomial distribution. However, the probability of a mutation is extremely low, and the number of nucleotide sites is vast. Under such conditions, the binomial distribution converges to a Poisson distribution (Cam, 1960).

Nevertheless, the Poisson distribution requires the variance to be equal to the mean, a condition not met by mutation count data. Due to the previously mentioned hotspot mutations, the observed mutation counts have a mean of 12.466 while the variance is 4228.057. This is called an overdispersion, *i.e.*, when the variance is greater than the mean (Coly et al., 2016). Therefore, negative binomial regression, which allows modelling this additional variability through an additional dispersion parameter (Coly et al., 2016), was chosen for further analysis, as it provides a more accurate and realistic representation of the observed mutation data.

In order to estimate the degree of mutational influence  $\beta_I$  using a negative binomial regression, the relationship described before was transformed by a logarithmic scale and added a parameter  $\alpha_0$  as the logarithm of the constant of proportionality, which captures the intercept for the regression. Final version used for the analysis can be seen below.

$$\log\left(\frac{\mathbb{E}[n(c, a)]}{f(c)}\right) = \alpha_0 + \beta_I * \log(\rho(c, a)) \quad (4)$$

$\beta_I$  represents the coefficient of mutational influence. Its value is expected to range from 0 to 1, with higher values indicating a stronger influence of mutation rates on the evolutionary outcome.

## 4. Analytical Methods and Results

To investigate the relationship between the structure of the genetic code and mutation frequency, we used three statistical approaches.

#### 4.1. Variance and Correlation Testing

In order to analyse the differences in the data, they were divided into groups according to the number of pathways through which a specific change is possible. The highest number of genetic code pathways leading to one amino acid exchange was 3, therefore there were 3 groups (See Figure 8). As the data were not normally distributed, the Kruskal-Wallis test was performed. The test showed no significant difference in the numbers of observed mutational events normalised by frequency of codons between groups ( $H = 1.928$ ,  $df = 2$ ,  $p = 0.381$ ).

A Spearman rank correlation test was performed according to the theoretical framework. The test evaluated whether there is a monotonic relationship between the number of distinct genetic code pathways from the original codon to the resulting amino acid and the number of mutational events observed for this specific codon-amino acid change, normalised by the genomic frequency of the codon. It did not reveal a significant association ( $\rho = 0.071$ ,  $p = 0.182$ ,  $n = 354$ ).

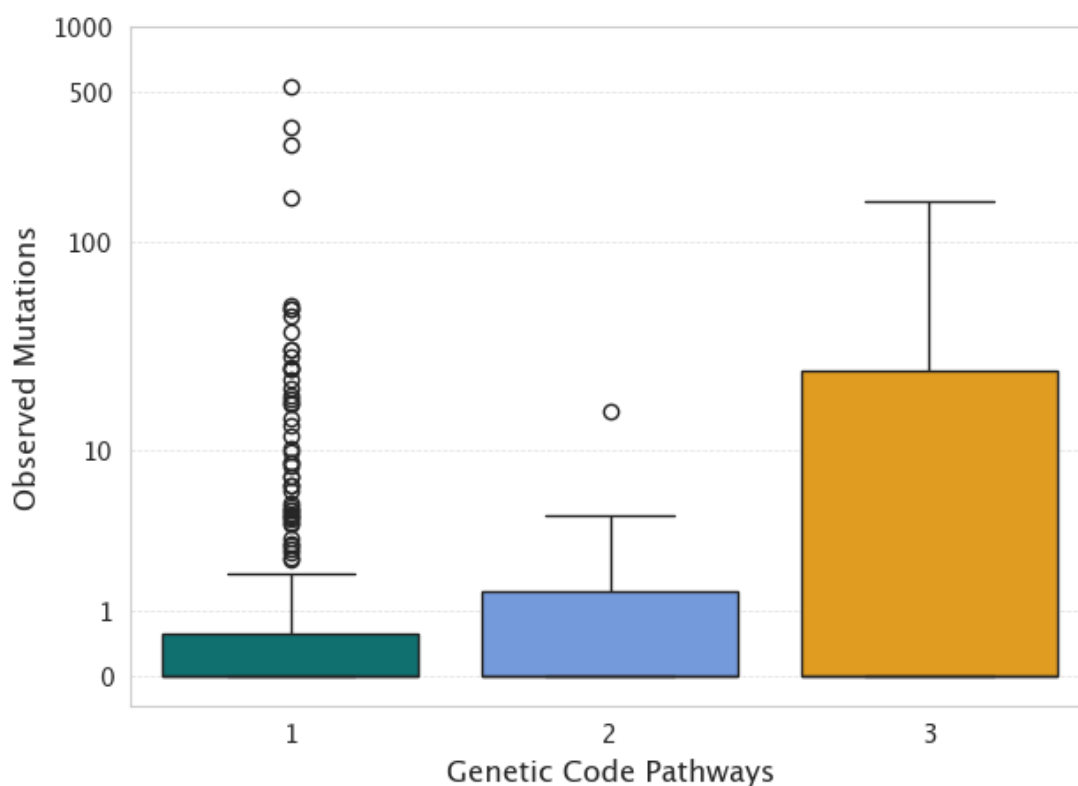


Figure 8. Distribution of observed mutations normalised by codon frequencies in *M. tuberculosis* grouped by the number of genetic code pathways. The y-axis is log-scaled to accommodate the highly dispersed distribution and outliers.

## 4.2. Regression Analysis

To effectively describe a relationship between variables, a predictive model must be constructed. Given the nature and distribution of the data, a generalised linear model with a negative binomial distribution was employed. The model was fitted to assess the relationship between the number of observed mutations, normalised by codon frequency, and the number of genetic code pathways according to Equation 4.

Regression was performed with the logarithmic transformation of the number of mutational paths as a predictor (See Figure 9). The estimated effect was positive but not statistically significant ( $\beta = 0.353$ ,  $SE = 0.252$ ,  $z = 1.402$ ,  $p = 0.161$ ,  $n = 354$ ). The model explained only a small proportion of the variance (Pseudo  $R^2 = 0.008$ ).

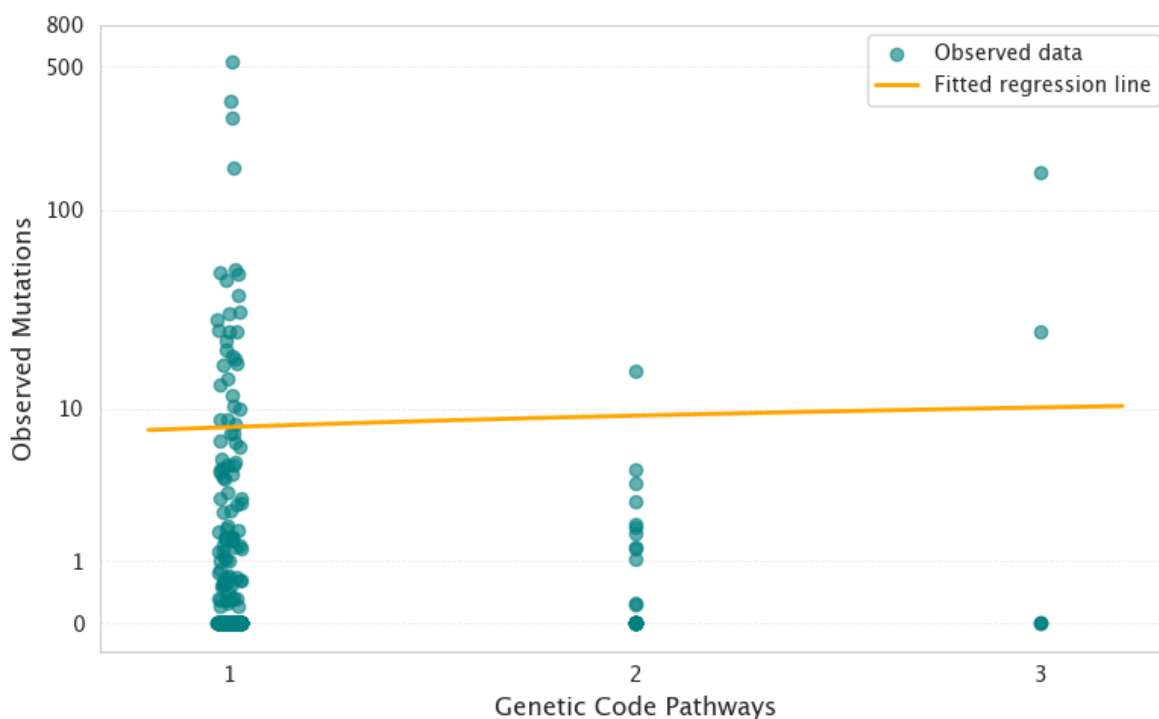


Figure 9. Relationship between the number of genetic code pathways and actual mutations for each change original codon – mutated amino acid. The observations, in teal, are normalised by the frequencies of the original codon from *M. tuberculosis* and depicted on a logarithmic scale. A negative binomial regression model was fitted to the log-transformed data, indicating a slight positive trend between the number of mutational paths and mutational frequency (in yellow).

## 5. Discussion

### 5.1. Summary of Findings

The structure of the canonical genetic code imposes constraints on amino acid changes possible through single-nucleotide mutations. Therefore, a mutation bias could favour amino acid substitutions that are accessible through multiple genetic code pathways over those accessible through fewer or only one. Nevertheless, the results of this study did not reveal a significant relationship between the observed mutation frequencies and the number of genetic code pathways.

The Kruskal-Wallis rank test revealed no significant differences in the number of observed mutations between groups with 1, 2, or 3 genetic code pathways. Likewise, Spearman's rank correlation did not show a significant association between the number of observed mutational events and the number of genetic code pathways. Finally, a generalised linear model with a negative binomial distribution did not identify a significant effect of genetic code pathways count on observed mutation frequency.

### 5.2. Interpretation

Several factors may explain the absence of a significant relationship in the results.

First, the reason why no mutation bias caused by the genetic code was detected could be the presence of a different mutation bias. For instance, as previously stated, transition bias is a widely occurring phenomenon in *M. tuberculosis*, in this particular dataset (Payne et al., 2019; See Figure 10). Any mutation bias could distort the distribution of observed mutations in a way that masks the expected effect of the number of mutational paths. Incorporating a transition/transversion distinction or other forms of mutation biases into the model could therefore be a useful next step.

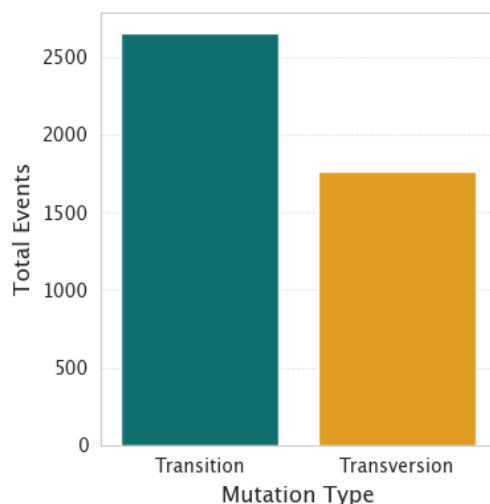


Figure 10. Transition bias observed in the dataset from Payne et al. (2019)

Second, as can be seen in Figure 11, using the example of codon AUG, the dataset was heavily influenced by mutational hotspots. Although the sample included 4413 observed single-nucleotide mutational events, the distribution was highly skewed, with a few mutations occurring at very high frequencies and many codon–amino acid combinations not observed at all. This lack of density may have weakened the statistical power of the tests and obscured the potential underlying patterns. The fact that observed mutations were concentrated in only a small subset of all possible codon-to-amino-acid changes reduced the overall predictive capacity of the model (Cano et al., 2022). Adding more observations to the dataset would help mitigate this effect by lowering the proportion of zero values and consequently, the relative importance of mutational hotspots. The Kruskal–Wallis and Spearman’s rank tests are both relatively robust to extreme values due to their reliance on ranks rather than absolute values. Although both of them were employed, this rank-based normalization proved insufficient in our case, given the high number of zero observations.

Third, the codon frequencies used to normalise mutation counts were calculated from whole genome data from *M. tuberculosis*. However, the observed mutations are limited to genes associated with antibiotic resistance, which may have a distinct codon usage bias compared to the entire genome. This discrepancy could have introduced noise into the normalisation step and weakened any potential signal.

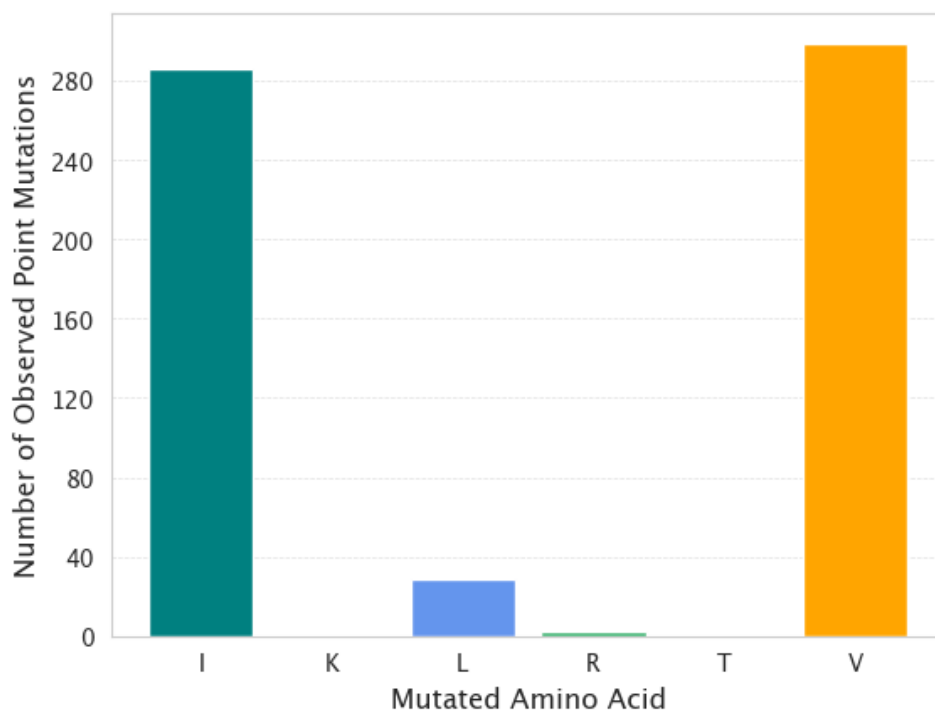


Figure 11. Counts of observed amino acid exchanges from methionine, codon AUG. According to our hypothesis, we would expect a ratio of 3 : 1 : 2 : 1 : 1 : 1 for isoleucine (I) : lysine (K) : leucine (L) : arginine (R) : threonine (T) : valine (V). However, we can see that hotspot mutations to isoleucine and valine disrupt the ratio.

Fourth, although the population-genetic conditions of *M. tuberculosis* likely align closely with origin-fixation dynamics, therefore the role of mutation bias could be significant, the number of possible mutational routes to antibiotic resistance is rather small (Cano et al., 2022). As mentioned above in chapter 1, section 3, a small mutational target size may limit the observable impact of mutation bias despite its presence (Cano et al., 2022). To overcome this limitation, it would be necessary to examine a different dataset, particularly one involving an alternative adaptive trait with a broader mutational target size.

Fifth, the model structure itself may have been overly simplistic. The relationship between mutational accessibility and observed mutation frequency may not be linear or even monotonic. Factors such as the structural context of the protein, local sequence effects, and epistatic interactions can all influence whether a mutation is tolerated or beneficial (Couce et al., 2015; Cano et al., 2023). A more complex model or a multivariate approach may be needed to account for these interactions.

### **5.3.Future Directions**

To address these limitations, future work could explore several directions. First, transition bias could be controlled either by explicitly including it as a variable in the analysis or by separating the data based on mutation type. Second, hotspot mutations could be analysed separately in order to assess their specific impact on the overall trends, or a wider dataset with less zero values and hotspot mutations could be tested. Third, codon frequencies could be recalculated using only genes associated with antibiotic resistance, rather than relying on whole genome frequencies, to better reflect the actual mutational context of the data. Additionally, future analyses could explore nonlinear modelling approaches or machine learning methods capable of capturing more complex relationships between variables. Finally, expanding the dataset to include a broader range of organisms or resistance contexts would improve the generalisability of the findings and increase the statistical power.

In conclusion, while the current study did not find a significant link between mutational accessibility and observed mutation frequency, it highlights the complexity of factors that influence adaptive molecular evolution. Further work is needed to disentangle these influences and better understand how the structure of the genetic code interacts with mutation bias, selection, and protein function to shape evolutionary outcomes.

# Conclusion

This thesis reviewed the concept of mutation bias, non-random tendencies in the types of mutations that arise, and examined its relevance in adaptive evolution. Various molecular, structural, and contextual sources of mutation bias have been identified, including transition and AT bias, CpG hotspots, sequence and structural context biases. Although traditionally mutation bias was considered a neutral background process, growing empirical evidence from studies on adaptive mutations shows that it can play a directional role in evolution. These findings suggest that mutation bias may shape evolutionary outcomes alongside natural selection, with implications for the predictability of evolution and applications such as improved predictions of resistance development, cancer progression, and other medically and environmentally significant evolutionary processes.

In the experimental part of this thesis, we investigated whether the number of genetic code pathways - that is, possible single-nucleotide changes, as constrained by the structure of the genetic code - influences the frequency of observed adaptive mutations. Using a dataset of mutations conferring antibiotic resistance from *M. tuberculosis*, we hypothesised that codon-to-amino-acid substitutions accessible through multiple genetic code pathways would be observed more frequently than those accessible through fewer pathways. However, the analyses performed in this thesis did not support this hypothesis. This result may reflect disturbing factors such as other mutation biases, the presence of mutational hotspots, limited target size, or insufficient statistical power due to dataset sparsity.

Altogether, these findings support the relevance of mutation bias in adaptation. Moreover, this thesis highlights that further investigation of mutation biases among adaptive mutations may lead to valuable insights, both in relation to well-characterised biases and to those that remain only hypothetical, such as a potential influence of the structure of the genetic code.

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